

**Prepared by:**

Andy Gomes

Approved by:

Clive Baek

Review Date: March 2026**Original Date:** September 2019

Safety Device Check Sheet Cleaver Brooks Boiler

This procedure assists in checking the safety devices on the boiler. These safety devices must be tested and calibrated on a Bi-Annual basis

Gas Pressure Switches

The high and low gas pressure switches need to be checked to confirm that they will function properly.

Step	Operation	Cutout Pressure Setting	Cutout Actual Pressure	Date Device Tested
1	Low Gas Pressure Switch			
2	High Gas Pressure Switch			

Pressure Control Safety Devices

The pressure is controlled in the boiler via the Weishaupt LMV controller. This boiler has a 10:1 turndown and the pressure modulates to maintain the header pressure. The Weishaupt controller will modulate to maintain pressure, and will shut down the boiler. This boiler is also equipped with two high pressure control devices. **To test the units first confirm the setting, and then lower down the setting to confirm the switch would trip the boiler. After the test is complete return the set point back to the original position.**

Operation	Pressure Setting (PSI)	Pressure which device trips	Date Tested
Test the first High Pressure Trip	135		
Test the second High Pressure Trip	140		

Water Control Safety Devices

This boiler houses water in a drum and the level is controlled by a level master controller. The controller has the following functions. Pump on/off control, low/high water cutoffs. The boiler is also fitted with a second water column with a float controlled manual LWCO. There is also probe fitted in the steam drum for LLWCO which is manual reset. The hot well is fitted with low water cutout probes which will turn off the boiler in the event of a low water condition.

Step	Operation	Date Tested
1	Low Water Control Level master	
2	High Water Control Level master	
3	Water Column Manual LWCO	
4	Steam Drum Manual LWCO	
5	Feed water Pump Capacity Test >160.5 PSI (Min 7% above Safety Valve Setting)	
6	Confirm Low Hot well Level Turns off Boiler	

Remote & Emergency Shutdown

As part of the ASA this plant can be remotely shutdown. The remote shutdown can be done from the BMS computer as well as the emergency shutdown button on the wall at CCM.

Step	Operation	Initial when Complete	Date and Time when Complete
1	Confirm the remote shutdown can be executed from the BMS computer.		
2	Confirm the boiler can be shutdown via the emergency shutdown button on the wall outside of the boiler room		